BCSE203E - Web Programming

Digital Assignment - II

Winter Semester 2025-2026

Topic: Develop Your Own JavaScript-Based Project- **SereniMind**

Student Details

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Assignment Submission Data

Below are the links to the source code repositories and the live deployment of the SereniMind project.

1. Main GitHub Repository (Official Classroom Repo)

- **Link:** https://github.com/JayakumarClassroom/daily-task-yoswinner
- **Contents:** Contains the `index.html`, `style.css`, and `script.js` files for SereniMind.

2. Coursework and Project Repository

This is my comprehensive personal repository containing all lab tasks, coursework, and the final iterations of the SereniMind project code.

- **Link:** https://github.com/yoswinner/Web-Programming-FS-25-26

3. GitHub Pages (Live Project Deployment)

The fully functional, live version of the SereniMind application, hosted via GitHub Pages for easy access and testing without needing a local server environment.

- **Link:** https://yoswinner.github.io/WebDev-Project-SereniMind/

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1. Project Proposal

Project Title: SereniMind — Student Mental Wellness Companion

Objective: The main objective of SereniMind is to provide students with a unified, accessible digital space to manage academic stress, improve concentration, and track their emotional well-being over time. As a student, I noticed how easy it is to experience burnout, so I wanted to build a practical tool that solves this problem directly in the browser—without requiring any app installations, sign-ups, or subscriptions. It's meant to be a quick, calming escape during long study sessions.

Features: 1. **Interactive Breathing Exercise:** A customizable, animated visual guide that helps users pace their deep breathing through Inhale, Hold, and Exhale phases.

2. **Pomodoro Focus Timer:** A built-in 25/5/15-minute productivity timer featuring a smooth SVG circular progress ring and gentle, programmatically generated 528 Hz audio notifications to signal breaks.

3. **Daily Mood Tracker:** An intuitive emoji-based logging system where users can record how they feel, add an optional text journal, and view a 7-day visual history to spot trends in their mood.

4. **Daily Affirmations:** A randomized motivational quote generator to give users a quick positive boost whenever they open the app.

5. **Responsive Glassmorphism UI:** A visually soothing, "aurora-themed" interface utilizing blurred background animations and smooth scroll-reveal interactions to create a relaxing environment.

Technologies: - **HTML5:** Used for creating a clean, semantic document structure, prioritizing accessibility and laying the groundwork for interactive elements like inline SVGs.

- **CSS3:** Heavily utilized for advanced keyframe animations, responsive Flexbox/Grid layouts, CSS Variables for theme management, and the modern "Glassmorphism" aesthetic using backdrop-filter.
- **Vanilla JavaScript (ES6+):** The core engine of the app. Used for extensive DOM manipulation, managing complex timers (setTimeout/setInterval), persisting data with the Web Storage API (localStorage), and triggering animations via the IntersectionObserver API.

2. Documented Steps

Development Process

Step 1: UI/UX Conceptualization and Setup I started by figuring out the "vibe" of the application. Since it's a mental wellness app, I decided on a calming, dark "aurora" aesthetic. I set up my CSS variables for a consistent color palette (deep purples, teals, and soft blues). To make it feel immersive, I created large, animated background blobs using CSS `filter: blur` and applied a glass-like frosted effect (`backdrop-filter`) to all the main content cards.

Step 2: Semantic HTML Structuring Before writing any complex logic, I mapped out the single-page application into distinct, semantic <section> blocks: Hero, Affirmation, Breathing, Focus, and Mood. I made sure to structure things logically so the JavaScript would be easier to write later. I also decided to hand-code an inline SVG for the Pomodoro timer's progress ring to keep the project self-contained without needing external image files.

Step 3: Implementing Core Interactions (JS) Next, I tackled the navigation. I built a sticky top bar and added a functional hamburger menu for mobile users. To make the page feel more dynamic, I used the JavaScript `IntersectionObserver` API. This watches as the user scrolls and smoothly fades in the different sections (`.reveal` classes) just as they enter the viewport.

Step 4: Building the Breathing Exercise This was one of the more fun parts to build. I wrote a recursive JavaScript loop utilizing nested setTimeout functions to handle the "Inhale -> Hold -> Exhale" cycle. I linked the timing of these JavaScript functions to the CSS transition durations, so the visual circle expands and contracts in perfect sync with the user's customized breathing pace.

Step 5: Developing the Pomodoro Timer

I used setInterval for the core countdown logic. To make the timer visually engaging, I implemented a dynamic formula to adjust the SVG `stroke-dashoffset`. As the timer ticks down, JavaScript updates this value, causing the circle to visually "drain." I also tapped into the browser's AudioContext (Web Audio API) to synthesize a soft 528Hz sine wave tone when a session completes, avoiding the need for clunky MP3 files.

Step 6: Creating the Mood Tracker & Local Storage Finally, I wanted users to be able to look back at their week. I built an interactive emoji grid for mood selection. When the user saves their mood, JavaScript packages the data (emoji, text, date) into an object, converts it to a string using `JSON.stringify`, and saves it to the browser's `localStorage`. When the page reloads, another function parses that data and dynamically generates a 7-day history grid and a weekly average score.

Challenges Faced & Solutions

1. **Challenge:** Handling overlapping timeouts in the Breathing Exercise. At first, if I rapidly clicked "Start" and "Pause," or changed the durations mid-breath, multiple timers would fire at once, causing the circle to glitch out wildly.
   - **Solution:** I realized I needed strict state management. I created a robust cleanup function (stopBreathing) that tracks all pending timeout IDs in an array. Now, whenever the exercise is paused or altered, it immediately calls clearTimeout() on everything in that array, ensuring a clean slate.
2. **Challenge:** Figuring out the math to smoothly animate the circular SVG Pomodoro timer.
   - **Solution:** After some trial and error, I learned I had to calculate the exact circumference of the SVG circle ($2 \times \pi \times r$) and apply this specific value to the stroke-dasharray in CSS. Then, I wrote a JS function that calculates the percentage of time remaining and updates the stroke-dashoffset to match, creating a perfect, smooth radial drain effect.
3. **Challenge:** Storing structured daily mood data permanently across browser sessions.
   - **Solution:** I knew I had to use localStorage, but it only accepts plain text strings, not JavaScript arrays or objects. To solve this, I learned how to serialize my structured mood data into JSON strings before saving them, and then parse them back into usable JavaScript objects whenever the page loads.
4. **Challenge:** Ensuring the app looked good on mobile devices. Initially, the large breathing circle and the Pomodoro timer would overflow and break the layout on smaller screens.
   - **Solution:** I heavily leaned into CSS Flexbox and media queries. I adjusted the scaling of the SVGs and reorganized the side-by-side buttons into stacked columns for screens under 768px wide, ensuring a seamless experience regardless of the device.

Resources & Tutorials Used

- **MDN Web Docs (Mozilla):** My primary reference for JavaScript syntax, deeply understanding how localStorage works, and figuring out the Web Audio API and IntersectionObserver.
- **CSS-Tricks.com:** I used their excellent guides on SVG Animations and Glassmorphism techniques to achieve the visual style I wanted.
- **JavaScript.info:** A great resource that helped clarify how the browser's event loop handles Promises and setTimeout/setInterval.
- **YouTube - Web Dev Simplified:** (e.g., *Learn LocalStorage in 12 Minutes*) Their concise tutorials really helped me understand how to structure my state and handle DOM updates cleanly without overcomplicating my code.
- **YouTube - Kevin Powell:** (e.g., *CSS Flexbox deep dives*) His tips on responsive design and modern CSS properties were absolute lifesavers when I was trying to fix my mobile layout issues.

3. Learning Reflections

Through the development of SereniMind, I went from knowing basic syntax to actually understanding how to piece a web app together. Here are my top 10 takeaways:

1. **Grasping Advanced DOM Manipulation:** I became much more comfortable dynamically creating, updating, and completely restructuring HTML elements and CSS classes on the fly using vanilla JavaScript.
2. **Mastering the Web Storage API:** I finally understand how to persist user data across sessions using localStorage, and why data serialization with JSON.stringify and JSON.parse is so crucial.
3. **Taming JavaScript Timing Events:** I learned the nuances between setInterval (great for a steady countdown clock) and complex, nested setTimeout loops (perfect for the phased steps of the breathing exercise).
4. **Performance with IntersectionObserver:** Instead of writing laggy scroll-event listeners, I learned how to use the Intersection Observer API to trigger beautiful, high-performance CSS animations exactly when elements enter the screen.
5. **Synthesizing Audio in the Browser:** I had no idea you could generate sounds purely with code! Using the Web Audio API to create oscillators and gain nodes was a huge lightbulb moment for me.
6. **Controlling SVGs via JavaScript:** I learned the mathematical relationship between stroke-dashoffset and stroke-dasharray, allowing me to build custom, animated radial progress bars from scratch.
7. **Modern CSS Aesthetics:** I significantly improved my design skills by combining backdrop-filter, CSS variables, and @keyframes to create a modern, "Glassmorphism" user interface.
8. **Event Delegation and Handling:** I learned how to efficiently attach event listeners to elements—even ones dynamically generated by JavaScript—and how to manage app state based on user interactions.
9. **True Responsive Web Design:** I stopped relying on fixed pixel widths and started effectively using Flexbox, relative units, and CSS media queries so the app fluidly adapts to phones, tablets, and desktops.
10. **State Management without Frameworks:** I learned how to centralize my timer and mood data into distinct JavaScript objects (like timerState). This taught me how to maintain a single "source of truth" in my app, which is a core concept used in big frameworks like React, but implemented here in pure Vanilla JS.

4. Importance of this Project

Written Reflection:

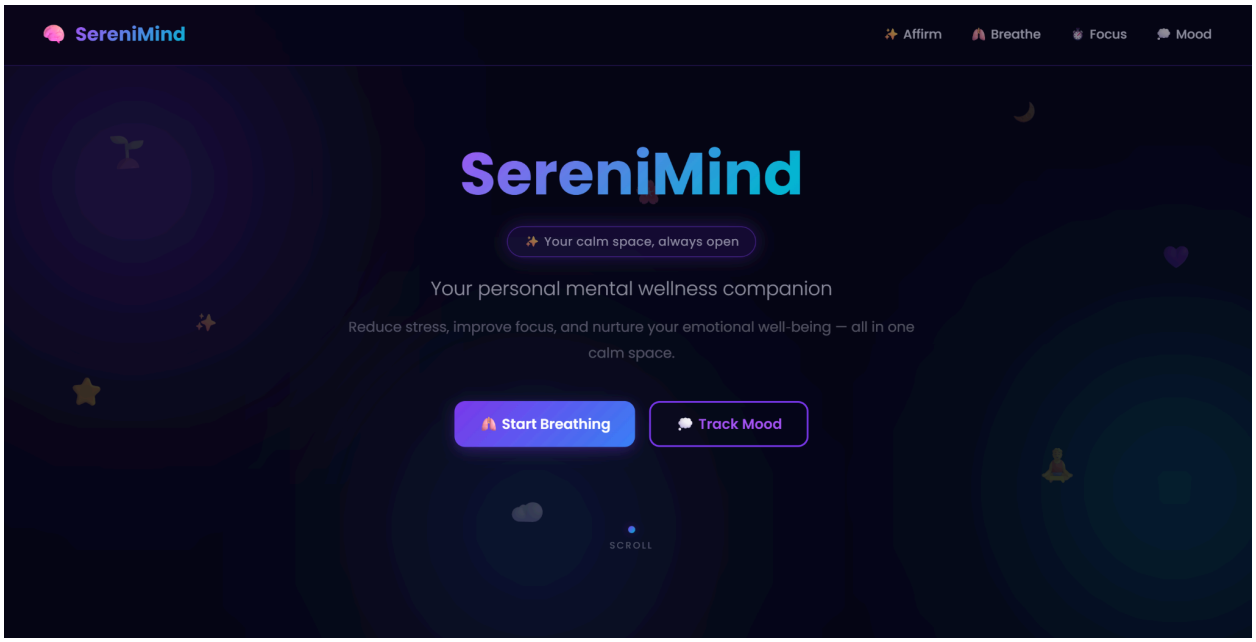
Mental health is a critical, yet frequently overlooked aspect of student life. Between rigorous academic schedules, digital assignments, and exam preparations, we constantly operate under high pressure. I've noticed myself and my peers experiencing severe stress, screen fatigue, and burnout.

The true importance of **SereniMind** lies in its immediate, frictionless accessibility. When you are overwhelmed, the last thing you want to do is download a new app, create an account, or navigate a paywall just to find a simple breathing exercise. By offering a web-based, completely free, and account-less platform, SereniMind completely lowers the barrier to entry for mental wellness.

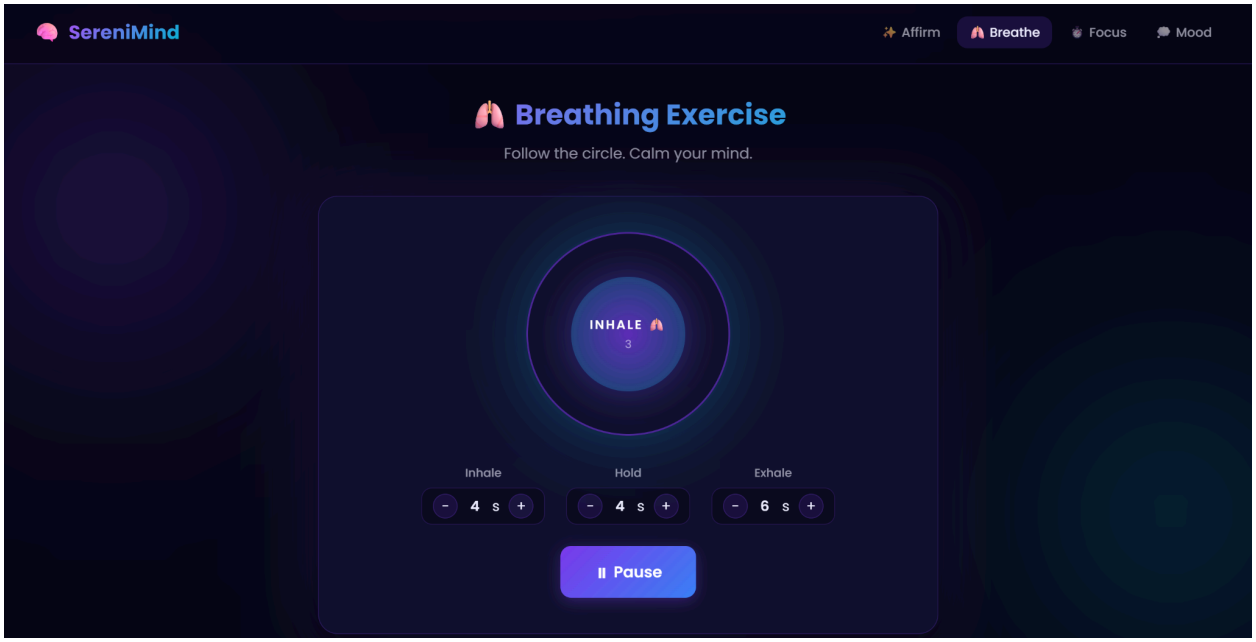
The features I chose are highly intentional and scientifically backed: the Pomodoro method prevents study fatigue by enforcing structured breaks; the guided breathing exercise physically lowers the heart rate and interrupts anxiety spirals; and the mood tracker encourages emotional self-reflection, helping users recognize patterns in their well-being over time. Building this project wasn't just an exercise in checking off JavaScript requirements; it allowed me to hone my technical skills while creating a genuinely practical, meaningful tool that I plan to continue using in my own daily routine.

5. Screenshots

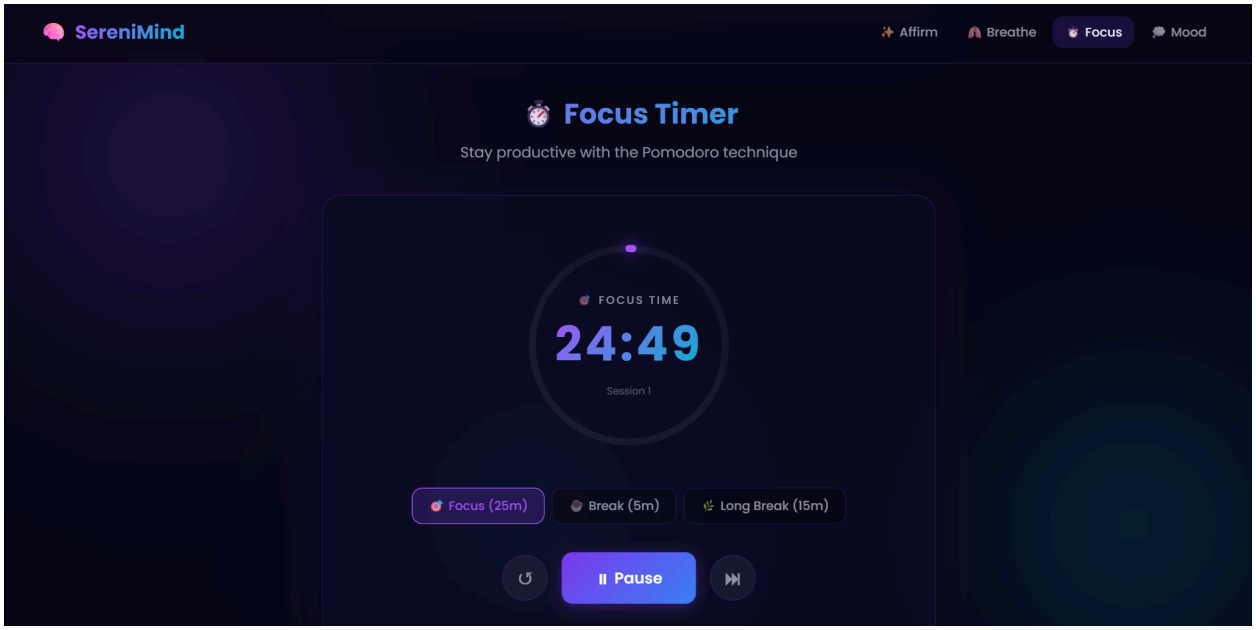
1. Hero Landing Page & Navigation:



2. Breathing Exercise Interface:



3. Pomodoro Focus Timer:



4. Daily Mood Tracker & History